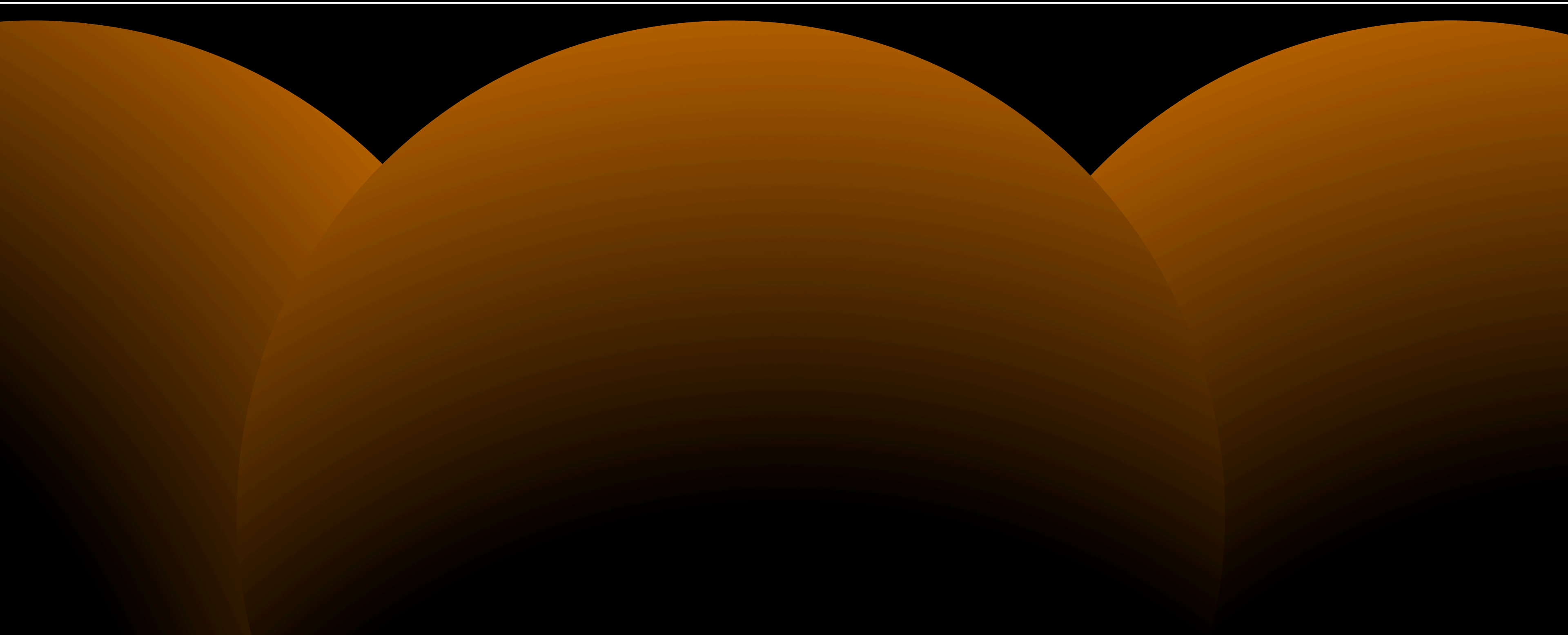


MINI SEARCH ENGINE

Using Data Structures and Algorithms



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Introduction

A Mini Search Engine is a software application that helps users search and retrieve information from a collection of documents efficiently. It uses Data Structures and Algorithms to organize and process data for faster searching. The system creates an index of keywords, allowing quick access to relevant documents based on user queries. By using techniques such as Hash Maps and Inverted Indexes, the search process becomes more efficient. This project demonstrates the practical application of DSA concepts in information retrieval. It provides accurate and faster search results while reducing search time. The Mini Search Engine is a simple model that helps understand how real-world search engines work.

Abstract

The Mini Search Engine is a software application designed to search and retrieve relevant information from a collection of documents efficiently. It uses Data Structures and Algorithms (DSA) such as HashMaps and Inverted Indexes to organize data and improve search speed. The system processes user queries, matches keywords with indexed documents, and displays accurate results quickly. This project demonstrates the practical application of DSA concepts in information retrieval. It reduces search time, improves efficiency, and provides a simple understanding of how modern search engines work.

Literature Survey

Several studies have shown that efficient information retrieval systems rely on Data Structures and Algorithms (DSA) to improve search performance. Techniques such as HashMaps and Inverted Indexes are widely used to organize keywords and enable faster document retrieval. Traditional search methods are slower for large datasets, whereas indexed search significantly reduces search time. Modern search engines use similar indexing and ranking techniques to provide accurate results. This project adopts these fundamental concepts to develop a simple and efficient Mini Search Engine for quick keyword-based document searching.

Problem Statement

- Searching through large amounts of data manually is slow.
- Traditional searching takes more time.
- Fast and efficient searching is required.
- The project aims to retrieve relevant information quickly using DSA.

Objectives

- Develop a simple search engine.
- Store and organize documents efficiently.
- Perform fast keyword searching.
- Display relevant search results.
- Understand real-world use of Data Structures.

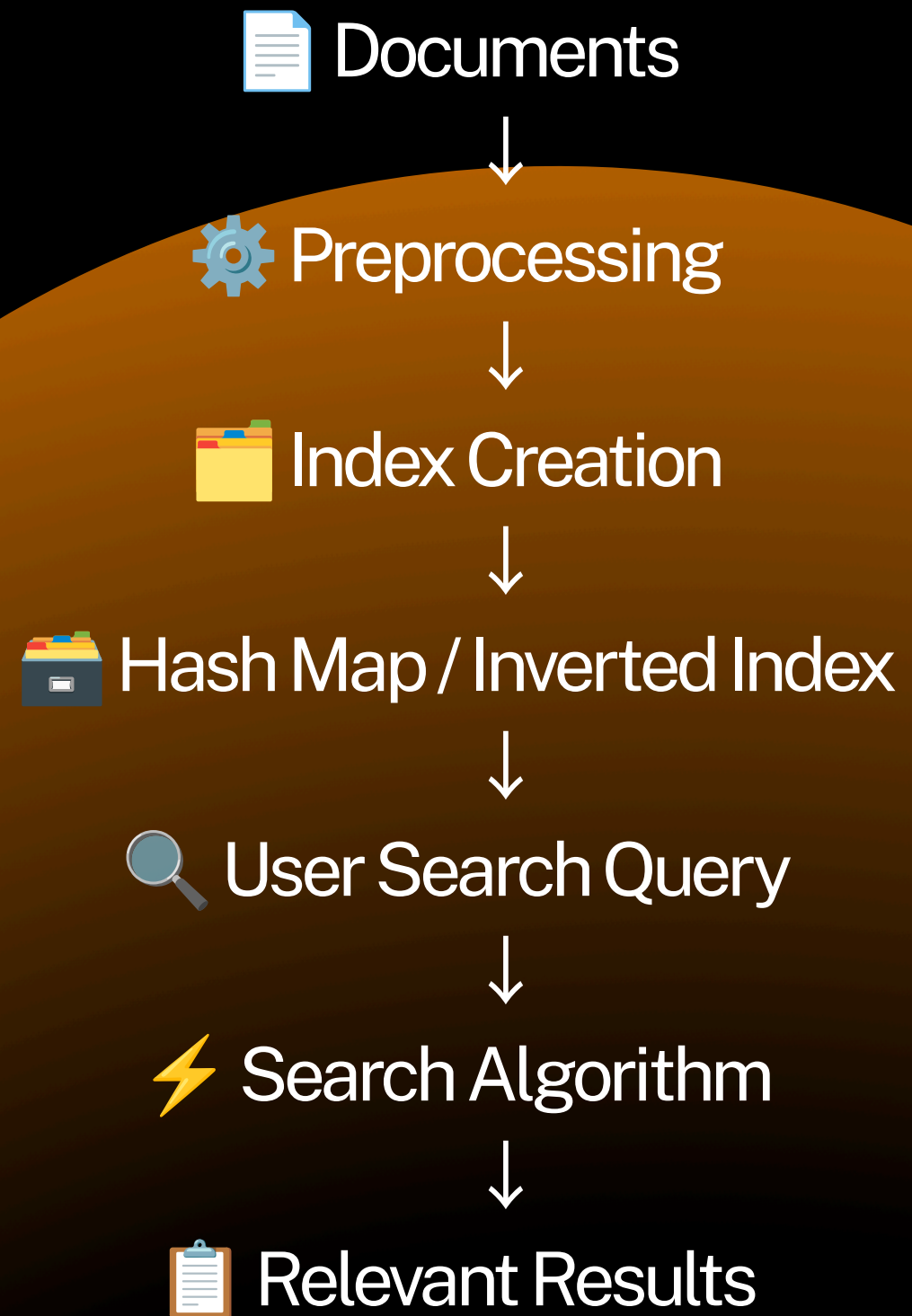
Data Structures Used

- Hash Map
- Array/List
- Inverted Index
- String Processing

Explain:

- Hash Map stores keywords and their document locations.
- Array/List stores document information.
- Inverted Index enables faster searching.

System Architecture



Methodology

1. Collect and store the documents in the system.
2. Preprocess the text by extracting keywords.
3. Create an inverted index using HashMaps for fast searching.
4. Accept the user's search query.
5. Match the query with the indexed keywords.
6. Display the relevant documents as search results.

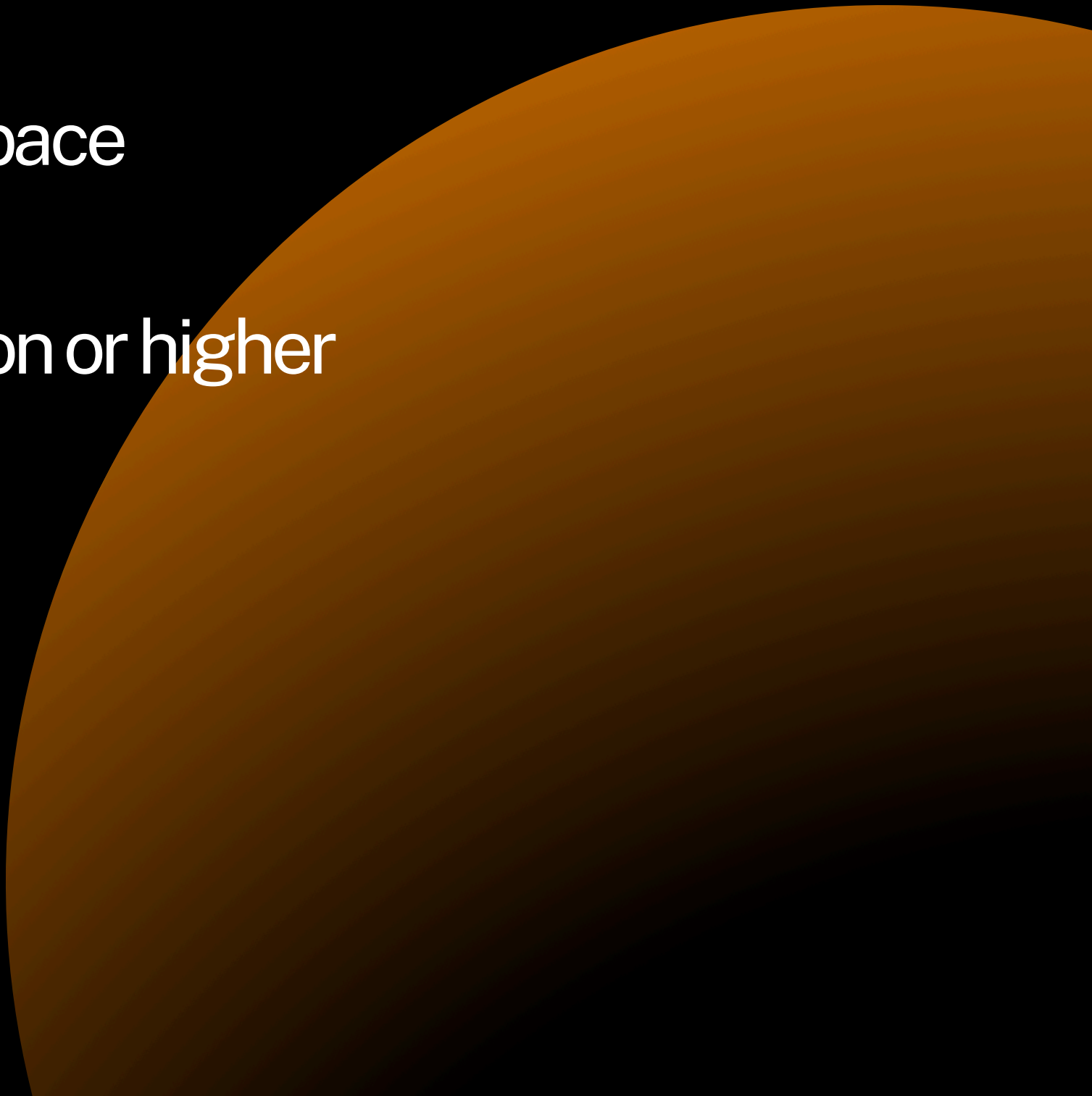
Algorithm

- Read all documents.
- Extract words.
- Build an inverted index.
- Accept the user's keyword.
- Search the index.
- Display matching documents.

Working

- User enters a keyword.
- The system searches the index.
- Matching documents are identified.
- Results are displayed instantly.

Hardware Requirements

- Processor: Intel Core i3 or above
 - RAM: 4 GB or higher
 - Storage: 500 MB free disk space
 - Keyboard and Mouse
 - Monitor: 1366 × 768 resolution or higher
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Software Requirements

- Operating System: Windows 10/11
 - Programming Language: Java
 - IDE: Eclipse / IntelliJ IDEA / VS Code
 - JDK: Java Development Kit (JDK 8 or above)
 - Database: Not Required (or MongoDB if your implementation uses it)
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Advantages

- Fast searching
- Efficient data retrieval
- Easy to implement
- Reduces search time
- Demonstrates practical DSA concepts

Applications

- Google Search
- Library Management Systems
- E-learning Platforms
- File Search Systems
- Digital Document Search

Future Enhancements

- Ranking of search results
- Spell correction
- Voice search
- Web crawling
- AI-based recommendations

Conclusion

The Mini Search Engine demonstrates how Data Structures and Algorithms improve search performance. By using indexing and efficient search techniques, the system retrieves relevant information quickly, making it suitable for real-world search applications.

The image features three overlapping semi-circles in a warm orange-brown hue, set against a solid black background. The semi-circles are arranged horizontally, with the central one being the largest and most prominent. The text 'THANKYOU' is centered within the largest semi-circle.

THANKYOU